

PET

Positron Emission Tomography, or PET, is an imaging technique that allows us to see how organs and tissues function inside the living body. It is based on the use of a radioactive substance, called a tracer, which emits tiny particles known as positrons. These tracers are usually molecules that participate in normal biological processes—such as glucose—so that their behavior in the body reflects real metabolic activity.

When the tracer is injected into the bloodstream, it travels to different parts of the body, accumulating in regions according to how active those tissues are. For example, in the case of fluorodeoxyglucose labelled with fluorine-18, areas that consume more glucose, such as the brain or a tumor, will collect more of the tracer.

After a short time, the radioactive atoms within the tracer begin to decay. Each decay releases a positron, which travels only a very short distance before it encounters an electron in the surrounding tissue. When a positron and an electron meet, they destroy one another in an event called annihilation, producing two photons of light that fly away in almost exactly opposite directions.

The PET scanner surrounds the patient with a circular array of detectors capable of sensing these pairs of photons. The system is designed to register only those pairs that arrive at two detectors at the same instant. When such a simultaneous detection occurs, the scanner knows that the annihilation—and therefore the source of the photons—must have been located somewhere along the straight line connecting those two detectors. By collecting millions of these coincidence events from all around the body, the computer reconstructs a detailed three-dimensional image (made of voxels) showing where the tracer is concentrated.

The brightness of each region in a PET image corresponds to how many annihilation events were detected there, which reflects the local biological activity of the tissue. Because of this, PET does not simply show anatomy—it reveals function: how cells are using energy, how blood is flowing, or how receptors are binding molecules.

Lesson 08/10/2025

With PET (positron emission tomography) we are able to get very specific and sensitive information, specifically targeting some compound of interest. In terms of brain activity, we can gather information about oxygen and glucose consumption, or blood flow supply or markers of synaptic activity. We can measure also other chemical compounds related to other chemical targets like neurotransmitters, not present in the brain tissue. This is possible knowing the metabolic fate of a tracer (^{18}F -Deoxyglucose).

PET imaging is important in determining neurodegenerative disorders. A possible valuation is a qualitative one, we primarily rely on static images, correlated to the concentration of a tracer, that is the number of annihilations. If we take in consideration glucose consumption, we can specifically consider this as a marker of neurons activity. ^{18}F -Deoxyglucose is better to use than anything else, for example water, because it is strictly related to the neuron, so that we recognize neurodegenerative disorders.

One of the painful aspects of dealing with the metabolic fate of the tracer is that these molecules are processed in different ways from different tissues. The brain takes about 45 minutes in order to have glucose equally distributed in all components, and for that time you must keep the brain

in the same mental state. Indeed, we can consider that as something that does not facilitate the investigation of mental function, whose time is much faster. Some tasks are very difficult to maintain for 45 minutes.

When you deal with aspects of synaptic activity that you like to refer to mental processes that are faster than the ones that you want to investigate, you can rely on markers that primarily are related to changes in blood flow supply, whose activity is much faster.

Example: H_2^{15}O half-life is 2 minutes, so in 10-15 minutes the activity of oxygen is disappeared, so you can monitor changes in blood flow that occur in few seconds. In some cases, oxygenated blood reaches a peak in the brain, that is compatible with several neural functions. Changes in blood flow supply are tightly coupled in changes in glucose and oxygen consumption (synaptic activities), so it can be used to represent synaptic activity, but you use tracers with different metabolic fate time.

When I'm dealing with an asymmetric distribution of glucose consumption and I'd like to compare these, you need to have absolute values for each of the voxels. Once you know the amount of dose that you have injected and the body mass of the participant, then you are able to reconstruct the curve, interpolating the values. Therefore, you are able to express Deoxyglucose as mg/100g tissue/minute (generally the range of value is 10-15 mg) or ml/100g tissue/minute (generally 50-70 ml).

Once you have the absolute value, we have the parametric or dynamic image, and at that point we can do any statistical operation with these images. When you take several volumes, you are trying to do an interpolation, so that you can have images to overlap. An important step is the spatial normalization, so that each we can move our brain into a standard reference state and have the possibility to compare different realizations (for example from different people). Each point is characterized by 3 coordinates. We use ATLAS (space of coordinates of reference), the first was the Talairach, obtained cutting the brain in slices and for many years it was the main reference for all brain studies.

In functional neuroimaging studies, cognitive subtraction refers to an aspect of experimental design involving the comparison of two conditions or brain states that are presumed to differ in only one discrete feature (the independent variable). Pure insertion asserts that there are no interactions among the cognitive components of a task

Donders' subtractive Methods

The main approach that we got derives from experimental psychology, that grew up from subtraction approaches.

Donders's work attempts to describe the processes going on in the mind, by analyzing cognitive activity into separate stages (many scientists assumed that responding to a stimulus occurred instantaneously!)

Donders was particularly interested in "timing the mind" and used a subtraction technique to time the different mental processes that the brain goes through when faced with different tasks.

Donders performed experiments using reaction time tasks in 1868:

1. a simple reaction time task - e.g. you are seated in front of a panel that contains a light bulb and a response button. When the light comes on, you must press the button. 190 ms